

Evaluating Damage to Vermont Bridges by Hurricane Irene with Multivariate Bridge Inspection and Stream Hydrogeologic Data

Ian Anderson¹; John P. Hanley²; Donna M. Rizzo³; Dryver R. Huston, M.ASCE⁴; and Mandar M. Dewoolkar, P.E., F.ASCE⁵

Abstract: The motivation for this study is the damage caused to over 300 long-span bridges in Vermont in 2011 by Tropical Storm Irene, which is part of the Hurricane Irene system. This study of the effects of this single extreme flooding event uses multiple variables related to these damaged bridges including bridge characteristics, stream characteristics, geographical features, environmental factors, and land-use resulting in a dataset with over 300 features. A new, evolutionary algorithm performed multivariate feature selection on this dataset to circumvent the computational challenges associated with using traditional statistical analysis. This impartial and exhaustive search of feature combinations generated nonlinear models that best predict bridge damage and identify bridges most at risk. Maps of statewide bridge vulnerability showed good correlation to damaged bridges. Features identified as significantly correlated with bridge damage include Irene stream power, and newly identified variables, such as watershed hydrologic soil types and the geographical context of specific watersheds. Channel rating and waterway adequacy rating, variables from the bridge inspection manual, also proved important. This research has applications beyond Vermont, as many of the newly identified variables can be created or monitored using commonly available data, and the analytical framework extends to other infrastructure assessment efforts at a system level. **DOI:** [10.1061/\(ASCE\)BE.1943-5592.0001603](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001603). © 2020 American Society of Civil Engineers.

Author keywords: Bridge damage; Extreme flood events; Tropical Storm Irene; Multivariate statistics; Feature selection.

Introduction

Extreme flood events cause damage to bridges throughout the world and substantial geomorphic change to the stream networks. Stream migration, degradation, aggradation, and widening are all natural processes as they attempt to maintain a dynamic *equilibrium* between the channel slope, stream discharge, the sediment load, and size (Lane, 1955). However, human encroachment and development in floodways have altered the natural stream course resulting in major conflict at the intersection of floodway infrastructure and geomorphic change. Bridges in particular are vulnerable to damage, as they are fixed nodes at the intersection of streams and human infrastructure, and susceptible to damage by changes in meander migration, channel width, depth, bank erosion, or embankment failure.

The majority of damage to short- and medium-span coastal bridges as a result of Hurricane Katrina in 2005 identified uplifting and hydrodynamic forces on the superstructure as the primary cause (Okeil and Cai 2008). A subsequent analysis of 262 bridges identified surge elevation as a key factor in determining the damage level from Hurricane Katrina (Padgett et al. 2012). Similar bridge infrastructure vulnerabilities have been witnessed at Escambia Bay, Florida, during the 2004 Hurricane Ivan (Douglass et al. 2004), and in Hokkaido, Japan, during the 2004 Songda Typhoon (Okada et al. 2006). More recently, severe flooding in September 2013 caused the collapse of 30 highway bridges and damage to an additional 20 bridges in Colorado (Kim et al. 2014).

The impetus for this study was the extreme flood event that occurred in Vermont in August 2011 as part of the Hurricane Irene system. Irene hit the Bahamas as a Category 3 hurricane, gradually weakening after crossing the Bahamas, and made landfall in North Carolina as a Category 1 hurricane that caused widespread damage across large portions of the eastern United States as it moved north-northeastward. The most severe impact of Irene occurred in the northeastern United States with catastrophic inland flooding in New Jersey, Massachusetts, and Vermont (Avila and Cangialosi 2013). In Vermont, the flooding impacted 225 of the state's 251 towns, with event rainfall totals of 100–200 mm (4–8 in.). Heavy rainfall at higher elevations caused flash flooding and progressed to widespread flooding throughout Southern and Central Vermont. Rainfall recurrence intervals exceeded 500 years in some towns and were in excess of 100 years in many areas. The storm damaged transportation infrastructure, as well as housing and businesses, over wide swaths of the state. Anderson et al. (2017a) identified 313 Vermont highway bridges (with spans greater than 6 m) as damaged in Tropical Storm Irene. Fig. 1 shows the location of all long structure (greater than 6 m) hydraulic bridges and their

¹Bituminous Concrete Manager, Vermont Agency of Transportation, 2178 Airport Rd. Unit B, Berlin, VT 05641 (corresponding author). ORCID: <https://orcid.org/0000-0001-7648-8987>. Email: Ian.Anderson@vermont.gov

²Former Graduate Research Assistant, Dept. of Civil and Environmental Engineering, Univ. of Vermont, 33 Colchester Ave., Burlington, VT 05405. Email: john.hanley21@gmail.com

³Professor, Dept. of Civil and Environmental Engineering, Univ. of Vermont, 33 Colchester Ave., Burlington, VT 05405. Email: drizzo@uvm.edu

⁴Professor, Dept. of Mechanical Engineering, Univ. of Vermont, 33 Colchester Ave., Burlington, VT 05405. Email: dryver.huston@uvm.edu

⁵Professor, Dept. of Civil and Environmental Engineering, Univ. of Vermont, 33 Colchester Ave., Burlington, VT 05405. Email: mdewoolk@uvm.edu

Note. This manuscript was submitted on July 19, 2019; approved on April 24, 2020; published online on August 6, 2020. Discussion period open until January 6, 2021; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Bridge Engineering*, © ASCE, ISSN 1084-0702.

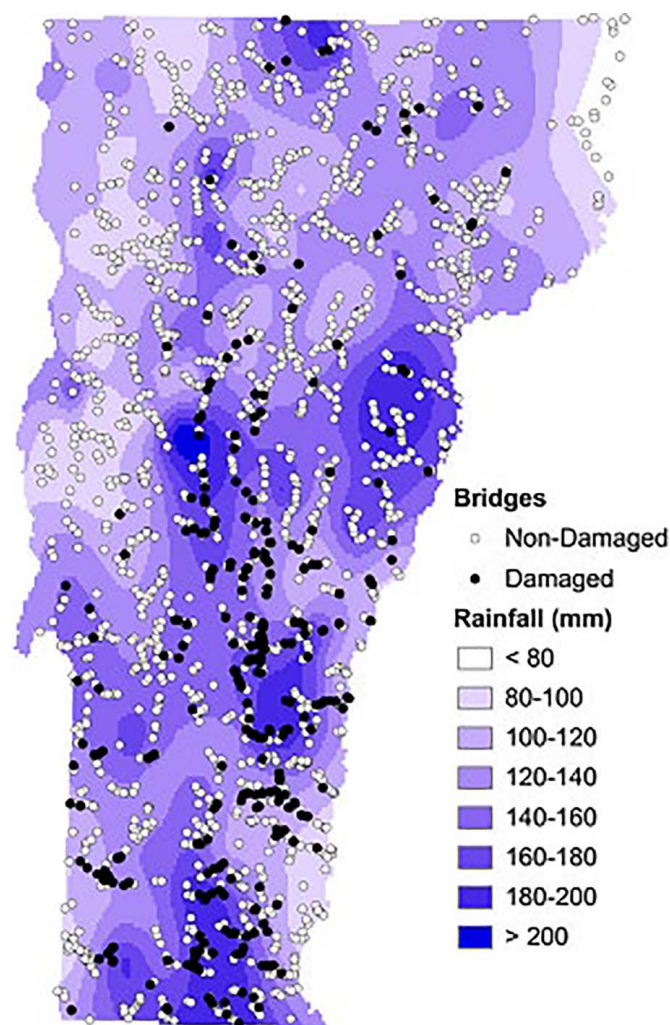


Fig. 1. Damaged and nondamaged bridges, over Tropical Storm Irene rainfall intensity.

damage state (313 damaged bridges and 1,936 bridges that did not have damage). These 313 damaged bridges comprised 55% steel beam, 34% concrete slab or beam, and the remaining 11% had historical steel or wood truss superstructures. Single-span bridges made up the vast majority, 82%, of bridges damaged, with 12% double span and the few remaining having three- and four-span structures. Anderson et al. (2017a) also reported that approximately 55.6% of the damaged bridges had scour damage, 29.7% had channel flanking, 8.3% had debris damage, and the remaining 6.3% had superstructure damage. Data were gathered on the estimated repair costs for 96 damaged bridges, and the average estimated cost of repair for scour, flanking, and superstructure damage was found to be about \$260,000, \$108,000, and \$18,000 per bridge, or \$318, \$120, and \$30 per square meter of deck area, respectively. Anderson et al. (2017a) assembled a comprehensive dataset of available bridge inspection and stream geomorphic assessments of Vermont hydraulic bridges that were damaged as well as not damaged in Irene.

Several factors affect bridges under extreme flooding events, including structural bridge characteristics, stream characteristics, geographical features, environmental factors, and land-use. This study builds upon the datasets assembled by Anderson et al. (2017a), adds new meaningful variables related to stream energy and watershed properties that were computed for all hydraulic bridges in Vermont (Anderson et al. 2017b), and then leverages this

enhanced dataset to identify those structural, geomorphic, geologic, and land-use characteristics (henceforth referred to as features) that are most correlated with damage to help identify bridges most at risk.

The aforementioned dataset assembled 330 features. Previous analyses of Vermont bridges damaged in Tropical Storm Irene (Anderson et al. 2017a, b) explored individual significance of the features associated with over 300 damaged and over 1,900 non-damaged bridges using multinomial logistic regression. The bridge age, specific stream power, event-based specific stream power, and various bridge ratings (e.g., channel rating, waterway adequacy rating, and scour rating) were individually found to be both statistically significant at discriminating between damaged and nondamaged bridges, as well as among different damage levels (categorized as slight, moderate, extensive, and complete damage). Although the aforementioned traditional statistical analyses provided very useful insight into the observed damage to over 300 bridges from a single extreme events, identifying sets of feature combinations that most closely correlate with bridge damage given hundreds of input variables is a severely challenging task for routine statistical techniques. For example, with 330 possible features, a four-effect model, using nominal logistic regression, would result in 5×10^8 possible feature combinations to explore. Commonly, the set of input features used in multiple regression analysis are reduced a priori using forward selection, backward elimination, stepwise selection, or expert opinion. Because it is not possible to explore all feature combinations exhaustively with traditional statistics, significant models (feature interactions) go undetected. In addition, the available features ranged in data type (i.e., nominal, categorical, and continuous) and had varying degrees of independence, or lack thereof. For example, the categorical variables such as the bridge type, foundation type, stream type, bed material, and other fields that may aid in the evaluation of bridge vulnerability could not be included; and some of the assembled features had missing values. For these reasons, the logistic regression analyses could employ only 210 out of 330 features.

To overcome these statistical limitations, we use a novel feature selection method to identify the combination of available features at play in complex bridge, stream, and watershed interactions. The method uses evolutionary algorithms (Hanley et al. 2020) to perform feature selection and eliminates the bias associated with a priori expert selection associated with really large data sets. The algorithm uses a nonparametric, multivariate approach to identify sets of multivariate features (i.e., the combined presence of bridge vulnerabilities, hydrogeological stressors, and increased flows) from a localized flooding event (i.e., Tropical Storm Irene) to better understand why bridge damage occurs. The feature combinations bring together variables from various sources that until now have not been utilized extensively in bridge assessments. Although in-depth analysis for individual bridges would be desirable, it is often impractical and too expensive at a regional scale when hundreds of structures are damaged in a single event. This paper explores such a system-level analysis of over 300 damaged bridges from a single extreme flood event using available information and uniformly generated information.

Methods

Data Collection

Anderson et al. (2017a) created a comprehensive dataset of bridge inspection and stream geomorphic assessments. Specifically, the following disparate sources of data were assembled and georeferenced into GIS (Geographic Information Systems):

1. All records of long structure bridges (greater than 6 m in span) and the inspection records to 2010 from the Vermont Agency of Transportation (VAOT) Bridge Inventory System (BIS). The Vermont state bridge inspections follow the NBIS (National Bridge Inventory System) criteria (FHWA 2015).
2. Estimates of poststorm damage level, damage type, and repair costs were obtained from the VAOT and the Vermont Department of Emergency Management (VDEM) documentation and supplemented using available inspection photos for all bridges affected. Bridge damage was categorized into four levels: slight, moderate, extensive, and complete, with damage descriptions provided in Table 1 and examples shown in Fig. 2. This damage ranking system was based on that proposed in HAZUS (Scawthorn et al. 2006) and later amended

Table 1. Description of damage categories used in the analysis

Damage category	Description
Slight ($n = 98$, 4.40%)	Channel erosion that does not affect the bridge foundation, superstructure and guardrail damage and debris accumulation without scour present
Moderate ($n = 123$, 5.52%)	Scour that affects the foundation, but not to a crucial state, bank and approach erosion, heavy aggradation and damage to the superstructure, but not to a crucial state
Extensive ($n = 42$, 1.88%)	Crucial scour, with some settlement to a single foundation, but not to the point of collapse, full flanking of both approaches, and superstructure damage that makes it structurally unsafe
Complete ($n = 50$, 2.24%)	Bridge washed away, collapsed, or has significant foundation damage that requires replacement

Source: Data from Anderson et al. (2017a).

Note: The number of bridges (n) in the dataset that corresponds to each damage category as well as the percentage out of all 2,229 bridges analyzed in this study is included in the table.

by Padgett et al. (2008). The ranking system descriptions were expanded to include the observed damage types, particularly damage from flooded river flow, including scour, flanking, debris, and superstructure damage.

3. Tropical Storm Irene rainfall data were collected for Vermont, the neighboring states of New York and New Hampshire, and the Province of Quebec. Rainfall and recurrence intervals were spatially interpolated using ordinary kriging to provide estimates over the entire state and enabled average rainfall and annual recurrence intervals to be calculated in each watershed.
4. The Vermont Agency of Natural Resources rapid geomorphic assessments (RGAs) provided a host of stream reach characteristics and measurements throughout the state. The RGA data characterize stream reaches, defined as river segments deemed consistent in slope, bed material, and condition (Kline et al. 2007). The RGA protocols are nationally recognized for providing a measure of stream disequilibrium and stream sensitivity indicating the likelihood of a stream responding via lateral and/or vertical adjustment to natural or human-induced watershed disturbances (Somerville and Pruitt 2004; Besaw et al. 2009). Joining the RGA data to the bridge inspection dataset linked the stream survey measurements and geomorphic metrics at the stream-reach scale directly to the bridge locations. Anderson et al. (2017b) extended the dataset reported in Anderson et al. (2017a) using additional watershed analysis:
5. An ArcGIS script was created to use a 10-m hydrologically corrected digital elevation model to iteratively delineate watersheds to the downstream end of each stream reach. Any rainfall runoff within a watershed will eventually flow to the outlet location. In total, 15,123 stream reaches were delineated to calculate their watershed area (i.e., the contributing area draining to some target location).

Stream power is the rate of energy (i.e., power) of the flowing water against the bed and banks of the river channel. Stream



Fig. 2. Bridge damage level and type from Tropical Storm Irene: (a) slight superstructure damage; (b) moderate debris damage; (c) extensive channel flanking damage; and (d) complete scour damage. (Images courtesy of Vermont Agency of Transportation.)

power functionally controls stream dynamics and morphology. Stream power estimates during extreme events show correlations to stream widening in Vermont (Buraas et al. 2014) and erosion and deposition in Colorado (Gartner et al. 2015). Key parameters that make up existing scour equations, such as flood duration and water velocity at each bridge site, would be strong predictors of bridge damage (Arneson 2013). However, these parameters were not available in existing databases or could not be efficiently computed for thousands of bridges individually. The lack of such data was the impetus for watershed-scale analysis and the use of stream power. The latter was both an available and a proven way of quantifying the stresses that flooding can impart on the streams and bridges.

The calculation of stream power in this analysis uses information widely available at a regional scale, rather than individually measured observations, to produce comprehensive estimates of stream power. A GIS script automated the estimation of stream power at any desired point, in this case at the locations of all hydraulic bridges in Vermont. Total stream power (Ω), also referred to as cross-sectional stream power (Fonstad 2003), is defined as follows:

$$\Omega = \gamma Q s \quad (1)$$

where γ = specific weight of water; Q = discharge; and s = energy slope. Specific stream power (ω) normalizes the total stream power by the width of the stream to estimate unit-bed-area stream power as follows:

$$\omega = \gamma Q s / b \quad (2)$$

where b = channel width. The discharge values required for stream power were calculated using regional regression equations for flood discharge at 50%, 20%, 10%, 2%, 1%, 0.5%, and 0.2% annual exceedance probability thresholds (Olson 2002, 2014). With the discharge at each target reach estimated, the energy slope was determined based on reach breaks established in the RGA protocols. Stream power, specific stream power (SSP), and scaled stream powers representing estimated Tropical Storm Irene intensities are described in Anderson et al. (2017b). Stream power is typically modeled on the 2-year flow, because higher flows often overtop the channel and dissipate energy along the floodplain. Thus, the stream power associated with a 2-year flow is considered to be more closely aligned with geomorphic change to the channel. The scaled stream powers, Irene stream power and Irene specific stream power, represent the full power within the channel of a contracted bridge, one with high embankments that force the full flow into the channel. Most bridge intersections in Vermont have raised embankments, built up over many years to avoid minor flooding, which creates a bottleneck of sort, forcing floodplain flows through the bridge opening, creating an intensified contraction effect.

Additional Data Generation

The following data were gathered or generated to supplement the aforementioned datasets as part of the work presented in this paper:

6. Land-use is an important metric in determining watershed characterization and understanding the relationship between rainfall and streamflow. Developed land cover contributes more storm-water runoff directly to surface water bodies, increasing flooding, while natural land cover and wetlands work to buffer flooding. The National Land Cover Database provided land-use classifications and was simplified from 16 types to 5 major groups for this study: developed, agriculture, open water/wetland, forest, and others. The hydrologic soil group (HSG), established by Mockus et al. (2009), is valuable for determining

runoff potential. The HSG is determined by the water transmitting soil layer with the lowest saturated hydraulic conductivity and is categorized into four distinct groups by runoff potential: A (low runoff), B (moderately low runoff), C (moderately high runoff), and D (high runoff). The percentage of each HSG for every stream-reach delineated watershed was calculated. Soil coverage with high runoff potential will contribute greater volumes of stormwater directly to rivers and streams, increasing the occurrence of flash flooding by shortening the time to peak flow.

Feature Selection Analysis Using a Conjunctive Clause Evolutionary Algorithm

In total, 330 features are available for every bridge in the combined dataset including the following: bridge inspection from the BIS, stream geomorphic data from the RGA, rainfall and recurrence interval, classification of the damage level and type, watershed analysis, and stream power metrics. An EA was applied to identify sets of key features from the 330 assembled features that best correlate to the bridge damage.

The EA of Hanley et al. (2020) is specifically designed for feature selection (i.e., identifying sets of multivariate interactions) associated with some desired outcome, k , of interest (e.g., level of bridge damage) associated within a highly large, complex dataset. The method searches across all multivariate combinations of input feature. Each feature may have a different data type (i.e., nominal, ordinal, and continuous). The algorithm evolves both the set of important features and the range of values using what are known as conjunctive clauses (CC) of the form

$$CC_k \text{ is defined as } F_i \in a_i \wedge F_j \in a_j \quad (3)$$

where F = feature (i.e., bridge or stream characteristic) whose value lies in the range a_i ; the symbol $\wedge$ = conjunction (i.e., features are linked by the logical AND statement). The EA selects CCs (set of features) that are probabilistically associated with some user-defined outcome, k (e.g., level of bridge damage) compared to what one might expect given the global distribution of k is equivalent to the positive predictive value (PPV) and is the ratio of true positive predictions to all positive predictions (actual damage detected to all predicted damage). Coverage, also called probability of detection (POD), is the ratio of true positive predictions to actual positive values (actual damage detected to total damaged). Order is defined as the total number of features in a given model (CC_k). Thus, a conjunctive clause with two features, each with a range at which it applies, joined by a logical AND operator is referred to as a second-order conjunctive clause. An example of a third-order conjunctive clause is: $CC = F_1$ (the stream reach has an out-flow elevation between 1,500 and 3,750 ft), AND, F_2 (has Soil B between 20% and 50%), AND, F_3 (stream power lies between 250 and 2,500 W/m). This conjunctive clause contains three distinct input variables, each joined by AND statements, and the range of values at which it applies. If this model were selected by the EA as an *optimal* solution, it would mean that all three variables (along with their evolved or reduced range of values) are highly correlated with a given outcome (in this case, level of bridge damage).

The algorithm processes in two phases, each using an age-layered population structure (Hornby 2006), and assesses fitness using a hypergeometric probability mass function (Kendall 1952) that accounts for the size of the dataset, the amount of missing data, and the distribution of outcome categories. The first phase evolves an archive of CCs consisting of feature combinations that have a high probability of a statistically significant association with a given outcome. The second phase evolves disjunctions of

these archived CCs to create an archive of probabilistically significant clauses in disjunctive normal form (DNF). The DNF outcomes are combinations of CCs, which in turn are combinations of features. The DNF order is defined by the number of CCs that comprise the DNF. Therefore, a second-order DNF consists of two conjunctive clauses joined by a logical OR operator.

The features identified as being important for discriminating bridge damage comprise all features archived in a second-order or higher conjunctive clause during at least one of the five repetitions. The main-effect features were only selected when that feature was archived across all five repetitions.

Multiple Logistic Regression

Following the EA feature selection analysis, the resulting feature combinations were tested with multiple logistic regression, conducted in the statistical software JMP 12 (version 12). The dependent variable was bridge damage at a moderate or greater level, with the feature combinations identified by the EA as the independent variables. To enable a direct comparison to the EA, logistic regression models were created to include the same variables identified in the EA feature combinations. Each logistic regression model was then used to create probabilistic estimates for prediction.

Results and Discussion

Evolutionary Algorithm and Logistic Regression

The EA identified feature combinations that statistically correlate with the observed bridge damage, combining features with AND statements. Damage levels classified as moderate, extensive, and complete were used as the positive state (damaged), while the two levels labeled nondamaged and slightly damaged comprise the negative state (nondamaged). The EA feature combinations are reported with their accuracy (PPV) and coverage (POD), two metrics commonly used for describing their quality. Feature combinations were identified (evolved) by the algorithm as good solutions based on the fitness function that considers both PPV and coverage. Fig. 3 plots the sets of feature combinations identified using the EA as points that show the tradeoffs between PPV and coverage. Of the 256 feature combinations stored by the EA, 29 input variables were identified by the EA. The range of variables, for each variable within a feature combination, was also optimized. The most selected variables and a list of seemingly important but not selected variables are shown in Table 2.

Fig. 3 shows that selected feature combinations vary by order and span a wide range of accuracies and coverages. Because there is a tradeoff between accuracy and coverage, no single solution is able to optimize both objectives simultaneously; in theory, the best solutions (ideal feature combinations) fall along what is known as a Pareto front (shown as the solid curve in Fig. 3).

Fig. 3 identifies one possible *optimal* solution (DNF 3313 represented by the hexagram) for predicting bridge damage. This solution comprises four sets of features. For illustration purposes, we highlight these feature combinations as pentagrams along a Pareto front on the same figure. Each feature combination is a conjunctive clause (CC) of unique model order. Table 3 identifies the variables and their range of values for each CC. The first-order model (pentagram associated with CC 5) has an accuracy and a coverage of 42.8 and 8.8, respectively. This feature includes waterway adequacy, which is a rating associated with the design capacity of the bridge for overtopping and shows what might be expected—that low ratings (0–3) are identified as being related to damage.

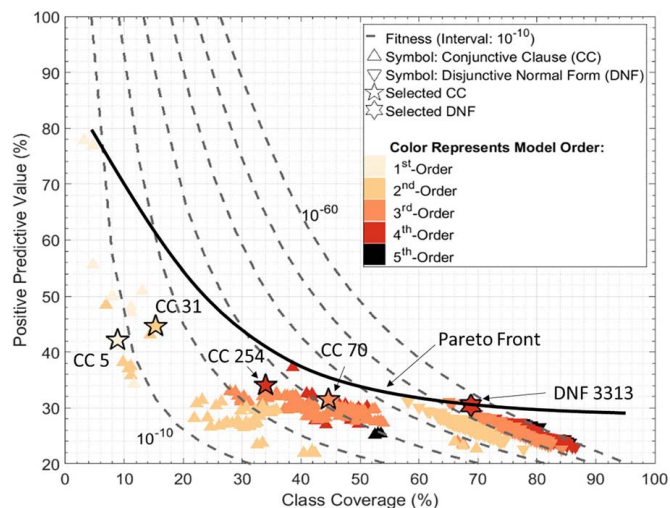


Fig. 3. Accuracy and coverage for conjunctive clauses (CC) and disjunctive normal forms (DNFs) of the conjunctive clauses. The contour lines represent equally spaced fitness using the hypergeometric PMF. Archived conjunctive clauses are represented by upward pointing triangles, and archived disjunctive normal forms are represented by downward pointing triangles. The axes represent the PPV and class coverage. The selected DNF which represents the fourth-order optimal disjunctive normal form is represented by a hexagram and the conjunctive clauses that comprise the selected DNF are represented by pentagrams. Darker colors represent higher orders of CCs and DNFs.

Table 2. Influential and noninfluential variables

Influential variables	Noninfluential variables
HSG B	Substructure rating
Subdominant watershed land-use	Superstructure rating
HSG C	Structure length
Stream power above	Span count
Irene stream power	Foundation type
Channel slope below	Deck rating
Elevation	Vertical clearance
Waterway adequacy	Structural adequacy and safety
Stream order	Serviceability and functional obsolescence
Land cover—water	Adjacent wetlands
Rainfall	Entrenchment ratio
Alignment of bridge deck and superstructure	Geomorphic degradation score
Land cover—development	Geomorphic aggradation score
HSG A	Valley slope
Irene specific stream power	Channel slope

The second-order feature combination, CC 31, has an accuracy and a coverage of 44.6 and 15.3, respectively, and identifies high Irene stream power and a low bridge inspection channel rating (prone to erosion) as being associated with damage. The third-order feature combination, CC 70, has an accuracy and a coverage of 29.7 and 52.6, respectively, and correlates to higher elevation, and high percent of Hydrologic soils B and C. The final feature combination, CC 254, is fourth order with an accuracy of 34.1 and a coverage of 34. This feature set includes high rainfall, low Hydrologic soil type D, lower than average open water land cover (less wetland and water bodies), and median-to-low confinement ratio (ratio of valley width to channel width). Each of the selected EA feature combinations is also analyzed using logistic regression, and the results are provided in confusion matrices (Tables 4–7). In

the confusion matrix, the upper-left-hand corner is the number of true positives; the upper right represents false negatives; the lower left shows false positives, and lower right represents the true negatives.

Table 3. Statewide CC feature combinations for DNF 3313

Conjunctive clause	Features and ranges
CC 5	DNF 3313—fourth order, POD = 27, PPV = 47.6 Waterway adequacy: 0–3
CC 31	Irene stream power: 8,000–12,500 W/m and Channel rating: 0–5
CC 70	Outflow elevation: 1,500–3,750 ft (457–1,143 m) Soil B: 20%–50% and Soil C: 45%–75%
CC 254	Rainfall: 5.25–7 in. (133.5–177.8 mm) Soil D: 0%–30% Land-use is water: 0%–5% and Confinement: 2%–14%

Table 4. CC 5 feature combination confusion matrices, EA, and logistic regression

CC 5 Ord 1 Actual	EA		Logistic regression	
	Predicted		Predicted	
	True	False	True	False
True	19	196	17	197
False	26	2,008	17	1,996

Table 5. CC 31 feature combination confusion matrices, EA, and logistic regression

CC 31 Ord 2 Actual	EA		Logistic regression	
	Predicted		Predicted	
	True	False	True	False
True	33	182	6	208
False	41	1,993	9	1,983

Table 6. CC 70 feature combination confusion matrices, EA, and logistic regression

CC 70 Ord 3 Actual	EA		Logistic regression	
	Predicted		Predicted	
	True	False	True	False
True	96	119	0	214
False	209	1,825	0	1,946

Table 7. CC 254 feature combination confusion matrices, EA, and logistic regression

CC 254 Ord 4 Actual	EA		Logistic regression	
	Predicted		Predicted	
	True	False	True	False
True	73	142	4	137
False	141	1,893	8	1,214

The EA feature combinations returned a higher number of true positives than the logistic regression, but also return a much higher number of false positives. Overall, logistic regression models using the same feature combinations identified by the EA produced fewer true and false positive results, indicating conservative statistical predictions that may limit its applicability for predicting bridge damage in practice. Logistic regression assumes that observations are independent, which is not necessarily true in this problem, given the spatial correlation between stream-reach locations. This independence issue is more challenging when feature selection is performed using either forward or backward substitution or expert examination. With hundreds of possible features, and numerous unknown interactions between features, the assumption of observation/feature independence is difficult. To avoid the independence and computational challenges often associated with traditional statistics on such large data sets, input data variables are often eliminated using expert judgement (i.e., domain experts preprocess the data and include only those variables deemed important). However, this greatly limits the power of large, comprehensive datasets.

Several variables identified by the EA highlight the value of methods capable of accommodating large numbers of input variables and the associated variable interactions. For example, Irene stream power is identified as a highly important indicator of bridge damage/vulnerability. Channel rating and waterway adequacy rating are two structural bridge inspection variables identified as being highly correlated with bridge damage, with low rating values being associated with high damage. High rainfall is included, which is expected, but still important to note. Hydrologic soil types are also identified as key variables, with higher percentages of B and C and lower percentages of D relating to bridge damage; and the number of available wetland or waterbodies (represented as open water land-use type) could act as storage to mediate the flooding. Lower percentages of open water land cover would mean a reduced storage area and potentially increased flooding.

The predictions of bridge damage from the EA and logistic regression deviate further from each other because higher order models are used. As one might expect, logistic regression produces results similar to the first-order EA models but fails to identify a meaningful number of damaged bridges for higher order combinations. In addition, the inclusion of multiple model effects in logistic regression produces exceedingly conservative results, failing to meet the required threshold to trigger a positive prediction.

Disjunctive Normal Form Analysis

The feature combinations identified by the EA show significant correlation to bridge damage, using only a small number of select bridge, stream, and watershed variables. The algorithm also simultaneously evolves the range of values associated with each of the selected variables, providing a threshold for discriminating between damage states. Damaged bridges can be correlated to a number of different variables, and combinations of variables, depending on the level of accuracy and coverage that the user deems most meaningful. The EA algorithm accommodates this type of analysis by creating sets of feature combinations that increase coverage while maximizing accuracy, in other words—joining CCs together with OR statements known as disjunctive normal forms (DNFs).

By joining sets of feature combinations associated with independent bridges, the DNF evolutionary algorithm produces results with better coverage than the individual feature combinations, while maintaining equivalent accuracy (Fig. 3 indicated with a hexagram). The selected set, DNF 3313, is fourth order, has an accuracy and a coverage of 27.6 and 68.4, respectively. The selected DNF comprises the individual feature combinations detailed in Table 3 and

highlighted in Fig. 3. Predicting damaged bridges based on DNF 3313 returned 148 true positives (70% of damage), 339 false positives (17% of nondamage), 67 false negatives, and 1,695 true negatives, and an odds ratio of 11. The odds ratio is the proportion of the true outcomes, divided by the false outcomes, and is a measure of association between exposure and outcome. The odds ratio represents the odds that damage will occur given the model conditions (select feature combinations and their associated range of values), compared to the odds of damage without them (data outside the selected feature combinations). DNF 3313 shows that bridges with parameters within the feature combination ranges are 11 times more likely to be damaged than those outside the group.

As a result, the DNF produced a model with higher coverage and similar accuracy, resulting in a greater number of predicted damaged bridges. Nearly 70% of the bridges damaged at a level classified as Moderate, Extensive, and Complete were successfully identified with 4 feature combinations comprised of 10 variables. Three of these variables, the channel rating, waterway adequacy rating, and Irene stream power, were identified as useful in determining bridge damage in the one-way analysis of Anderson et al. (2017a, b). Irene stream power is the scaled version of stream power, meant to represent the true flood stress passing through a constricted bridge, and is shown to be an important metric in damage. Channel rating assesses the condition of the channel and bridge riprap present during the biennial bridge inspection. Waterway adequacy is the rating of the ability of the bridge to convey its design storm flow, or the hydraulic overtopping flow associated with the service level of the bridge. Highway bridges are designed to convey larger recurrence interval flood events without overtopping, whereas rural town bridges are often designed for lower flow events. The percentages of hydrologic soil types within each bridge's watershed drainage area, identified by including the expansive set of watershed variables found through the watershed analysis, have proven important in correlating to bridge damage. High percentages of Hydrologic soil types B and C (each above the average) correlate with increased damage. Soil C is associated with increased runoffs and could precipitate flooding. Additionally, an alternative feature combination included a lower prevalence of Soil D, which is normally associated with higher runoff. The features that paired with a given soil type combinations may indicate distinct relationships between locations with higher and lower runoff potential that both lead to bridge damage. When the land cover proportion of *water* type is low, it is more significantly related to bridge damage, as are higher elevations. Higher elevations are the source of stream headwaters, have generally higher slopes, and are more prone to flash flooding. When using the EA algorithm to select DNF sets, we identify bridges with high Irene stream power and rainfall, poor channel and waterway adequacy ratings, higher elevation, little open water, and higher percent Soil types B and C to be at greater risk for flooding damage.

It is interesting to note that the EA identified features such as the rainfall, Irene stream power, waterway adequacy rating, channel rating, elevation, percentage HSG, open water land cover, and confinement ratio that predicted damaged bridges well. Structural features such as structure type, materials, scour rating, and year built, which one would expect to be correlated to damage, were not identified as adding value in the prediction models. Earlier analysis, a Kruskal–Wallis one-way analysis of variance (ANOVA), found the individual variables year built and scour rating correlated strongly with damage (Anderson et al. 2017a). Structural features not being identified through this selection indicate that they do not add to the prediction of observed damage in these multi-feature models, or more likely, they are correlated to and covered by other variables that had better prediction value. Although the scour rating

did have individual correlation to damage, it was a weak predictor and likely was replaced by channel rating and waterway adequacy as stronger predictors that both correlate with the scour rating valuable. The EA selects variables solely based on their prediction capacity, thus, eliminating the need for a priori selection (or reduction) of variables.

Geographic Prediction

The prediction of the bridge damage can be visualized geographically to highlight spatial relationships. The feature combinations associated with DNF 3313 (Fig. 3) allow for the prediction of bridge damage to be superimposed on major watersheds throughout the state of Vermont (Fig. 4). The map depicts distinct geographical areas where the bridges are predominantly predicted as damaged. The majority of bridges classified correctly as damaged were located in the central region of the state, following areas of high rainfall, such as the Otter Creek, the Connecticut-Waitsfield River to White River, and Connecticut-White River to Bellows Falls. However, the majority of the damaged bridges in the Winoochi watershed were misclassified as nondamaged. The White River watershed, which had a significant amount of damage, had a mix of both correct and missed damage predictions, with the missed predictions being grouped in tributary watersheds. The Mad River sub-watershed received a high amount of rainfall and experienced intense flooding, but the bridge damage was not correlated to the variables found in DNF 3313. The difference between bridge identification in neighboring watersheds is an interesting one and suggests that additional variables are needed to discriminate damage

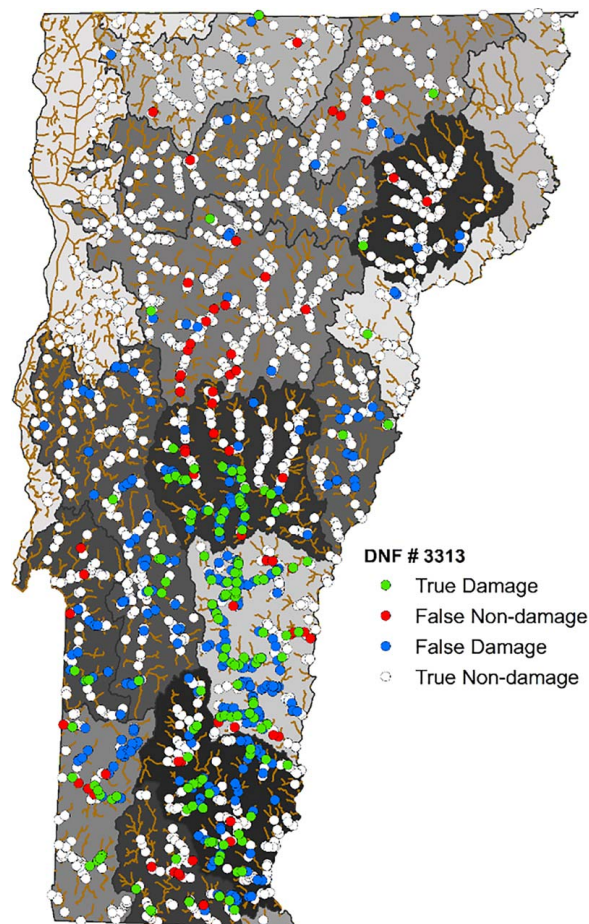


Fig. 4. Spatial relationship of DNF 3313 results.

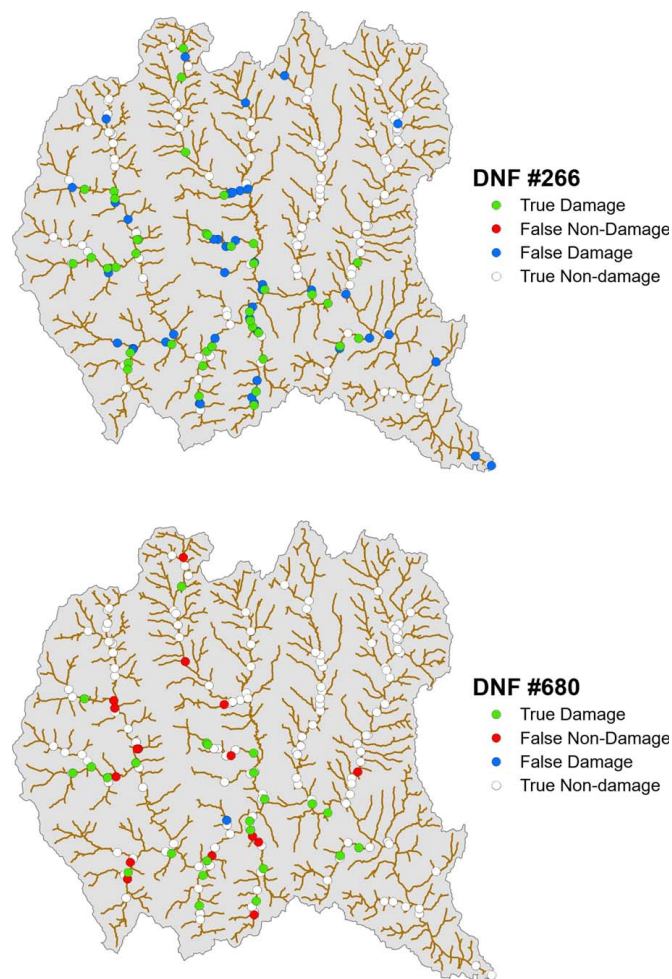
Table 8. White River watershed CC feature combinations for DNFs 266 and 680

Conjunctive clause	Features and ranges
DNF 266—fourth order, POD = 100, PPV = 47.6	
CC 278	AADT: 0–8,000 Federal sufficiency rating: 100–400 and Channel rating: 4–5
CC 309	AADT: 1,000–2,000 and Channel rating: 5
CC 677	Stream power up: 80–316 W/m Stream power down: 50–200 W/m and Structural evaluation: 0–5 and Waterway adequacy: 0–6
CC 971	Irene specific stream power: 250–2,500 W/m Soil B: 10%–45% and Waterway adequacy: 5–7 and Type of wearing surface: timber and asphalt
DNF 680—third order, POD = 60, PPV = 96	
CC 128	Soil B: 5%–25%
CC 718	Stream power up: 1,000–2,500 W/m Year built: 1930–1960 and State sufficiency rating: 0–400
CC 949	Irene specific stream power: 250–2,500 W/m Soil B: 15%–25% Waterway adequacy: 5–8 and Type of wearing surface: timber and asphalt

across watersheds and geographic areas. The results suggest that the analysis is identifying differences on a watershed scale, while being applied at the statewide level. The EA could potentially be used to identify a different DNF solution that may be useful in damage detection in those misidentified areas.

Single Watershed Prediction

Following the statewide prediction, and the observation of differences that seem related to geographic regions, an analysis was conducted on a single major watershed. The White River watershed was selected, with a total of 180 bridges, 40 of which had been Moderately, Extensively, or Completely damaged. Two DNFs were selected from this analysis, to help show the versatility of the method, and the ranges of prediction available. The first selection, DNF 266, was the result with maximum coverage (POD), correctly identifying 100% of the damaged bridges, which also results in overprediction of damage. The second, DNF 680, was the result with maximum accuracy (PPV), which limits the falsely predicted damage to 1 bridge, resulting in fewer bridges being correctly identified as damaged. The additional variables that make up the conjunctive clauses of DNFs 266 and 680 (Table 8) result in the best performing models. Bridge damage predictions from DNFs 266 and 680 are geographically represented in Fig. 5. For DNF 266, a number of the false damage predictions occur near true damaged bridges, as expected given their proximity and the spatial extent of some variables used to make the prediction.

**Fig. 5.** White river watershed feature selection predictions.

Implications for Practice

The work performed here could be used by bridge and river engineers to identify structures and rivers vulnerable to damage under extreme events. Transportation planning professionals dealing with resiliency could utilize this work to understand the diffuse risk within a network to allow for key infrastructure to be protected. By understanding and identifying the vulnerability in the system, agencies can target improvements and ensure redundancy, so all critical destinations remain accessible. Individual watersheds or sets of watersheds can be analyzed with the selection algorithm to aid in the updation of resiliency models. Rainfall models could be simulated to see how damage predictions react, and which bridges are identified as at risk.

Conclusions

This study used a unique dataset of over 300 long-span Vermont bridges damaged in 2011 Tropical Storm Irene, which was part of the Hurricane Irene system. Multiple variables related to these damaged bridges in this single extreme flooding event including bridge characteristics, stream characteristics, geographical features, environmental factors, and land-use resulted in a dataset with over 300 features. The dataset also included these features for the remaining 1,936 long-span Vermont bridges that were not damaged in the flooding event. The evolutionary algorithm utilized in this study to aid in bridge damage prediction combines features together, and sets of features to improve detection, and is capable of handling the large datasets brought

to bear in this study. As a result, the predicted damage can be used to assess the critical needs of the transportation networks and identify vulnerable links during extreme flood events.

The EA algorithm created a prediction set that included a significant number of true positive solutions, using the relatively small set of variables: rainfall, Irene stream power, waterway adequacy rating, channel rating, elevation, percentage HSG, open water land cover, and confinement ratio. The EA outperformed the artificially seeded traditional logistic regression (when provided with the same information); as the latter is overly conservative, failing to classify affirmative results, and requiring an assumption of independence between samples, which is likely violated given the spatial correlations between bridges.

When viewed from a geographic perspective, the prediction of damage using the optimal DNF appears to segment different watersheds. The difference between adjacent watersheds reveals the spatial relationship behind some of the parameters and displays how these differences affect the outcome, bridge damage. The EA watershed-scale analysis generated predictions of bridge damage that closely match the actual observations. Model solutions can be selected to create predictions of ranging conservatism, in order to capture all damaged bridges, or ensure over matching is excluded.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository or online in accordance with funder data retention policies:

- Database: <http://www.uvm.edu/~mdewoolk/?Page=form/form.php>; and
- EA: <https://www.mathworks.com/matlabcentral/fileexchange/69950-ccea-and-dnfea>.

Acknowledgments

This work was funded by the Vermont Agency of Transportation (VAOT) and the U.S. DOT through the University of Vermont Transportation Research Center. Partial support from Vermont EPSCoR with funds from National Science Foundation Grant EPS-1101317 is also acknowledged. The views and findings stated here are those of the authors and do not reflect the views and findings of the agencies mentioned. The authors are grateful to the VAOT Structures and Hydraulics sections, VANR, VDEM, Milone and MacBroom Inc., and Vermont Land Trust for providing some of the resources for this work, particularly to Nick Wark, Pam Thurber, Carolyn Carlson, James McCarthy, Christopher Benda, Roy Schiff, Evan Fitzgerald, and Jessica Louisos. Suggestions made by Alan Howard and Drs. Jeff Frolik, Scott Hamshaw, and Kristen Underwood are also greatly appreciated.

Supplemental Materials

Tables S1 and S2 are available online in the ASCE Library (www.ascelibrary.org).

References

- Anderson, I., D. M. Rizzo, D. R. Huston, and M. M. Dewoolkar. 2017a. "Analysis of bridge and stream conditions of over 300 Vermont bridges damaged in Tropical Storm Irene." *Struct. Infrastruct. Eng.* 13 (11): 1437–1450. <https://doi.org/10.1080/15732479.2017.1285329>.

- Anderson, I., D. M. Rizzo, D. R. Huston, and M. M. Dewoolkar. 2017b. "Stream power application for bridge-damage probability mapping based on empirical evidence from Tropical Storm Irene." *J. Bridge Eng.* 22 (5): 05017001. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001022](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001022).
- Arneson, L. A. 2013. *Evaluating scour at bridges*. FHWA-HIF-12-003. Washington, DC: Federal Highway Administration.
- Avila, L. A., and J. Cangialosi. 2013. "Tropical cyclone report hurricane Irene 21–28 August 2011." National Hurricane Center, NOAA. Accessed February 24, 2019. https://www.nhc.noaa.gov/data/tcr/AL092011_Irene.pdf.
- Besaw, L. E., D. M. Rizzo, M. Kline, K. L. Underwood, J. J. Doris, L. A. Morrissey, and K. Pelletier. 2009. "Stream classification using hierarchical artificial neural networks: A fluvial hazard management tool." *J. Hydrol.* 373 (1–2): 34–43. <https://doi.org/10.1016/j.jhydrol.2009.04.007>.
- Buraas, E. M., C. E. Renshaw, F. J. Magilligan, and W. B. Dade. 2014. "Impact of reach geometry on stream channel sensitivity to extreme floods." *Earth Surf. Processes Landforms* 39 (13): 1778–1789. <https://doi.org/10.1002/esp.3562>.
- Douglass, S. L., S. Hughes, S. Rogers, and Q. Chen. 2004. *The impact of Hurricane Ivan on the coastal roads of Florida and Alabama: A preliminary report*. Mobile, AL: Coastal Transportation Engineering Research and Education Center, Univ. of South Alabama.
- Fonstad, M. A. 2003. "Spatial variation in the power of mountain streams in the Sangre de Cristo Mountains, New Mexico." *Geomorphology* 55 (1–4): 75–96. [https://doi.org/10.1016/S0169-555X\(03\)00133-8](https://doi.org/10.1016/S0169-555X(03)00133-8).
- FHWA (Federal Highway Administration). 2015. Recording and coding guide for the structure inventory and appraisal of the nation's bridges. Rep. No. FHWA-PD-96-01. Washington, DC: US Dept. of Transportation.
- Gartner, J. D., W. B. Dade, C. E. Renshaw, F. J. Magilligan, and E. M. Buraas. 2015. "Gradients in stream power influence lateral and downstream sediment flux in floods." *Geology* 43 (11): 983–986. <https://doi.org/10.1130/G36969.1>.
- Hanley, J. P., D. M. Rizzo, J. S. Buzas, and M. J. Eppstein. 2020. "A tandem evolutionary algorithm for identifying causal rules from complex data." *Evol. Comput.* 28 (1): 87–114. https://doi.org/10.1162/evco_a_00252.
- Hornby, G. S. 2006. "ALPS: The age-layered population structure for reducing the problem of premature convergence." In *Proc., 8th Annual Conf. on Genetic and Evolutionary Computation*, 815–822. New York: ACM Press.
- Kendall, M. G. 1952. Vol. 1 of *The advanced theory of statistics*. 5th ed. New York: Hafner.
- Kim, Y. J., W. Marshall, and I. Pal. 2014. "Assessment of infrastructure devastated by extreme floods: A case study from Colorado, USA." *Proc. Inst. Civ. Eng.* 167 (4): 186–191. <https://doi.org/10.1680/cien.14.00032>.
- Kline, M., C. Alexander, and S. Pytlík. 2007. *Vermont stream geomorphic assessment protocol handbooks*. Waterbury, VT: VAO-N.
- Lane, E. W. 1955. "The importance of fluvial morphology in hydraulic engineering." *Proc. Am. Soc. Civil Eng.* 81 (795): 1–17.
- Mockus, V., J. Werner, D. E. Woodward, R. Nielsen, R. Dobos, A. Hjelmfelt, and C. C. Hoelt. 2009. "Part 630 Hydrology National Engineering Handbook." Chap. 7 in *Hydrologic soil groups*. Washington, DC: US Department of Agriculture, Natural Resources Conservation Service.
- Okada, S., H. Mitamura, H. Ishikawa, and S. R. Hokkaido. 2006. "The collapse mechanism and the temporary restoration of Omori Bridge damaged by the storm surge of Typhoon No. 18 in 2004." *Tech. Memo. Public Works Res. Inst.* 40 (9): 185–192.
- Okeil, A. M., and C. S. Cai. 2008. "Survey of short- and medium-span bridge damage induced by Hurricane Katrina." *J. Bridge Eng.* 13 (4): 377–387. [https://doi.org/10.1061/\(ASCE\)1084-0702\(2008\)13:4\(377\)](https://doi.org/10.1061/(ASCE)1084-0702(2008)13:4(377)).
- Olson, S. 2002. *Flow-frequency characteristics of Vermont streams*. Water-Resources Investigations Rep. 02-4238. Reston, VA: US Geologic Survey.
- Olson, S. 2014. *Estimation of flood discharges at selected annual exceedance probabilities for unregulated, rural streams in Vermont, with a section on*

- Vermont regional skew regression*, edited by A. G. Veilleux. Scientific Investigations Rep. 2014-5078. Reston, VA: US Geologic Survey.
- Padgett, J., R. DesRoches, B. Nielson, M. Yashinsky, O. S. Kwon, N. Burdette, and E. Tavera. 2008. "Bridge damage and repair costs from Hurricane Katrina." *J. Bridge Eng.* 13 (1): 6–14. [https://doi.org/10.1061/\(ASCE\)1084-0702\(2008\)13:1\(6\)](https://doi.org/10.1061/(ASCE)1084-0702(2008)13:1(6)).
- Padgett, J. E., A. Spiller, and C. Arnold. 2012. "Statistical analysis of coastal bridge vulnerability based on empirical evidence from Hurricane Katrina." *Struct. Infrastruct. Eng.* 8 (6): 595–605. <https://doi.org/10.1080/15732470902855343>.
- Scawthorn, C., et al. 2006. "HAZUS-MH Flood loss estimation methodology. II. Damage and loss assessment." *Nat. Hazard. Rev.* 7 (2): 72–81. [https://doi.org/10.1061/\(ASCE\)1527-6988\(2006\)7:2\(72\)](https://doi.org/10.1061/(ASCE)1527-6988(2006)7:2(72)).
- Somerville, D. E., and B. A. Pruitt. 2004. *Physical stream assessment: A review of selected protocols for Use in the clean water Act section 404 program*. Prepared for the U.S. Environmental Protection Agency, Office of Wetlands, Oceans, and Watersheds, Wetlands Division (Order No. 3W-0503- NATX). Document No. EPA 843-S-12-002. Washington, DC: U.S. Environmental Protection Agency.